

TRIADS — managing change in a complex world

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The telecommunications industry is characterised by enormous complexity and rapid change. The TRIADS project was set up to support business-wide architectural design and planning in this environment. A system has been developed to provide management and visualisation of BT's architectural framework, together with impact analysis facilities and support for the migration planning process. This paper outlines the business model underpinning TRIADS and describes some of the functionality of the system.

1. Introduction

With the drive to provide new and improved products and services, the networks and management systems deployed by telecommunications operators are becoming increasingly complex. In such an environment, change management becomes very difficult and it is essential to have a strong architectural framework to facilitate high-level design and migration planning [1]. This framework must take a broad view of the company's operations and encompass all the key dimensions appropriate to change management, from high-level business drivers, through process modelling and systems design, to technical deployment.

As the rate of change increases, the architectural planning process must itself adapt to keep pace with business needs. The *ad hoc* production of weighty documents is no longer appropriate. Furthermore, as interdependencies between different parts of the organisation become stronger, it is inappropriate for individual units to work in isolation. To facilitate rapid change, architectural information must be business-wide, dynamically maintained and provided on-line to all parties who require it.

The telecommunications requirements, impact analysis and decision support (TRIADS) project has addressed these needs by bringing together architectural information from diverse areas of BT and introducing a common data repository with multi-user access. The architectural information is accessed through a powerful front-end application which enables the data to be displayed and edited from multiple, customised viewpoints within a user-friendly graphical environment. The application adds significant value to the data by providing a range of facilities for navigation, information processing and impact analysis, together with

electronic tool support for the architectural planning process.

This paper outlines the scope of the TRIADS database and describes the methodologies employed in TRIADS to manage and view architectural information. Details of the functionality provided by the TRIADS application are presented.

2. The TRIADS business model

Change within BT is driven by a range of factors, including strategic initiatives, market pressures, competitor activity, regulation and the economic environment. In response to these drivers, programmes of development work are typically set up to deliver one or more of the following:

- new or improved products and services,
- re-engineered business processes,
- new or enhanced management systems,
- improved networks which exploit new technologies.

These programmes are often highly inter-dependent. For example, the provision of a new service may require changes to several management systems and the introduction of a number of new network capabilities.

The management of change is clearly a complex task. Any business model which is going to help tackle the complexity must be able to handle the large number of architectural dimensions involved and hold data on the relationships between them. Fig 1 shows a highly simplified schematic of the model employed in the TRIADS database.

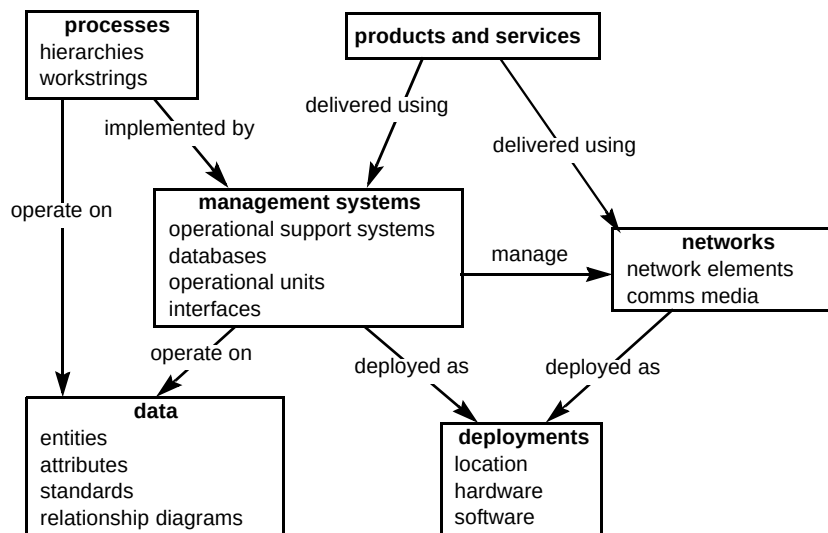


Fig 1 Schematic of the TRIADS business model.

The business model is based on six major architectural domains:

- products and services,
- processes,
- data,
- management systems,
- networks,
- deployments.

In addition, all major components of the model are time dependent and there are linkages to the programmes responsible for managing changes.

The TRIADS project has focused on development of the process, data and management system domains. The model deals primarily with generic relationships between versions of systems and network elements, rather than with deployed instances at specific sites. On moving into the deployments domain the volume of data expands rapidly and it is more difficult to justify the effort required to populate and maintain the data. The products and services and networks domains have already been modelled in other databases (for example, the product source database and the network capability register) and progress in these domains is likely to involve building bridges to the existing models.

The TRIADS model contains a detailed set of relationships between the various domains. For example, it is possible to identify which operational support systems (OSS) or operational units implement which business processes (for a particular service), which processes operate on which data entities, or which physical data entities are stored on which databases.

3. Viewing TRIADS information

3.1 TRIADS diagrams

TRIADS information is typically viewed using topological diagrams in which each node and link is bound to an instance in the database (see Fig 2). These diagrams provide a flexible means of displaying a wide range of architectural viewpoints. For example, a workstring may be represented by a set of process nodes connected by data flows, or a systems view may be represented by a set of OSS, databases, people, etc, connected by manual or electronic interfaces.

Each diagram is constructed from a header instance, a set of defining instances and a set of content instances. The diagram header holds basic details of the diagram, such as its name, date and diagram template. (The diagram template determines which entity types are permitted in the diagram and which symbols should be used to display them.) The diagram header is linked to the set of instances which define the diagram and the set of instances which are contained in the diagram. The content links hold display details such as co-ordinates and line styles which can be used to save diagram layouts to the database.

Basic attributes of the content instances (for example, name) are shown on the TRIADS diagram. Full details of an instance can be obtained by 'clicking' on the appropriate node or link and displaying the instance form. The form displays both the attributes of the instance and its relationships to other instances.

3.2 Navigating around the information

The construction of TRIADS diagrams facilitates diagram-to-diagram navigation using both cross-reference and drill-down methods.

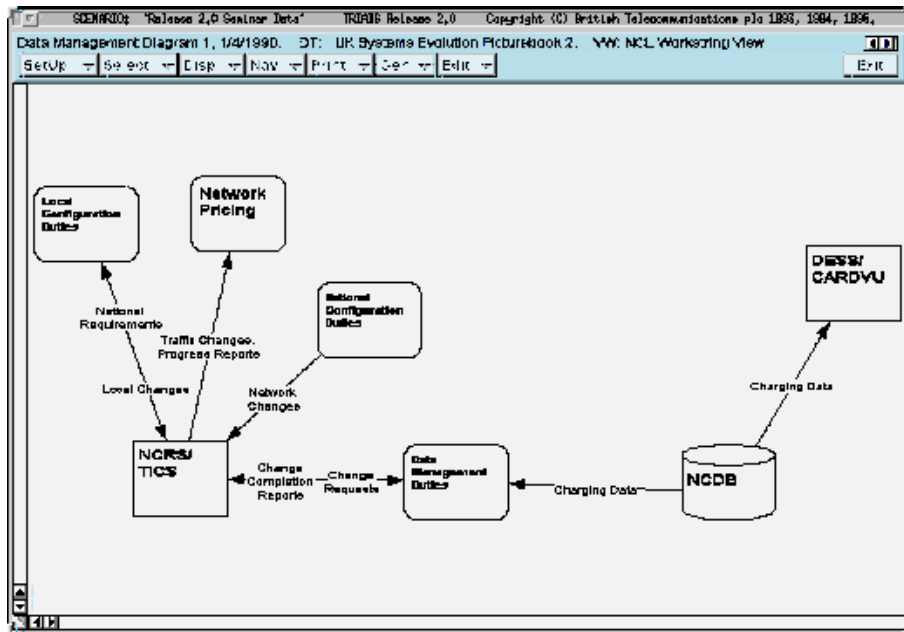


Fig 2 A typical TRIADS diagram.

Each real-world entity (for example, a particular version of an OSS) is typically modelled by a single instance in the TRIADS database and this instance is associated with each diagram node representing the entity. A simple search of diagram contents can therefore be used to find all diagrammatic occurrences of the instance and enable the user to cross-reference appropriate diagrams. Using this approach, an information trail can be followed through a succession of related diagrams. On navigating to a new diagram, the node or link used for cross-reference is highlighted to indicate the entry point for the trail.

Navigation through complex architectures is often further facilitated by the use of top-level diagrams and drill-down. A top-level diagram provides an abstraction of the problem domain which can readily be assimilated. Any particular aspect of the problem domain can be expanded by 'clicking' on the appropriate node and 'drilling down' to a lower level diagram. Drill-down is enabled by the use of defining instances for TRIADS diagrams. Any diagram which is defined by a particular instance may be accessed by drilling down from a node representing that instance in a higher level diagram. Multiple drill-down routes are therefore supported. For example, drilling down on a process node may provide access to both a lower level workstring diagram and a set of systems diagrams showing implementation of the process for various products. The TRIADS drill-down method is fully recursive, enabling a problem domain to be broken down into multiple levels of abstraction.

3.3 Creating diagrams

TRIADS supports both manual and automatic methods for creating diagrams. As the size of the database has

grown, there has been a shift in emphasis towards populating fundamental relationships and using auto-generation methods to create transient, customised viewpoints on the underlying model. TRIADS has thus been instrumental in bringing about a major cultural change within the business from diagram-based work practices to a much more powerful model-based approach.

Diagrams can be created manually using a simple 'drag and drop' approach. The symbol set displayed on the drag and drop palette is determined by the diagram template specified in the diagram header. Each symbol corresponds to a single entity type (process, database, network element, etc). A symbol is first positioned on the diagram and then 'populated' to bind it to an instance in the database. A list of suitable instances for population is presented to the user to encourage instance reuse and avoid population inconsistencies (for example, avoiding the population of a node with an OSS which is not due to become operational until after the specified diagram date). If the required instance does not exist it can be created by filling in a new form. A similar approach is employed to add and populate links between diagram nodes. Having added and populated all required nodes and links, a range of tools can be applied to improve the layout and appearance of the diagram. For example, symbols can be aligned and annotations can be added. The completed diagram can be saved to the database.

Diagrams can be automatically generated using either the standard viewpoint generation tools (for example, the hierarchy tool, the information flows tool and the timeline tool), or the auto-layout routines provided for the creation of fully customised viewpoints. Further details are presented in sections 4–6 below.

4. Customised information processing and data visualisation

4.1 The analyser

The most powerful feature of the TRIADS application is its ability to perform customised information processing and data visualisation. These features are provided by the TRIADS 'analyser'. The analyser enables users to gain maximum value from the data stored in TRIADS by fully exploiting the relationships in the business model. Analyser capabilities range from basic 'query tool' functionality, such as databases searching and custom report generation, to complex information processing and autogeneration of diagrams.

The analyser is essentially a powerful database search and information processing tool. The principles governing its operation are summarised in Fig 3. Requests are submitted to the analyser in the form of predefined scripts. These scripts may be created by the user or selected from a script library. The scripts are written in a high-level macro language which hides the complexity of the database schema and employs terms which are familiar to business architects. Output from the analyser may consist of a list of data items, a formatted report or a TRIADS diagram. This output can be directed to a screen, to a printer, or to a file for export to other software packages (for example, word processors and spreadsheets).

The more powerful features of the analyser may be grouped into three main categories:

- error checking — facilities range from identification of duplicate data instances to consistency checking of whole architectures,

- impact analysis — customised searches enable all instances affected by a change to be identified,
- viewpoint auto-generation — results of a database search may be viewed as a diagram, enabling users to create customised viewpoints on the business; the layout of nodes and links is calculated using 'force directed placement theory' [2].

Applications of the analyser are widespread and far-reaching. Three examples of the analyser's use are provided in the following sections. Further examples may be viewed on the WorldWideWeb [3].

4.2 Generating customised viewpoints

The analyser enables the user to view TRIADS data from whatever perspective is most appropriate to meet a particular need. For example, a user may wish to view the set of interconnected systems which is used to implement a particular service at a particular date. A simple analyser script could be written to carry out the following operations:

- search the database for all systems (OSS, databases, people, etc) that are related to the specified service and that exist at the specified date,
- search the database for all interconnections between the systems identified in the first step,
- display a diagram showing all the systems and interconnections identified in the first and second steps.

Auto-generated viewpoints have the advantage of always being entirely consistent with the model held in the database. However, when necessary, these diagrams may be saved to the database in just the same way as manually

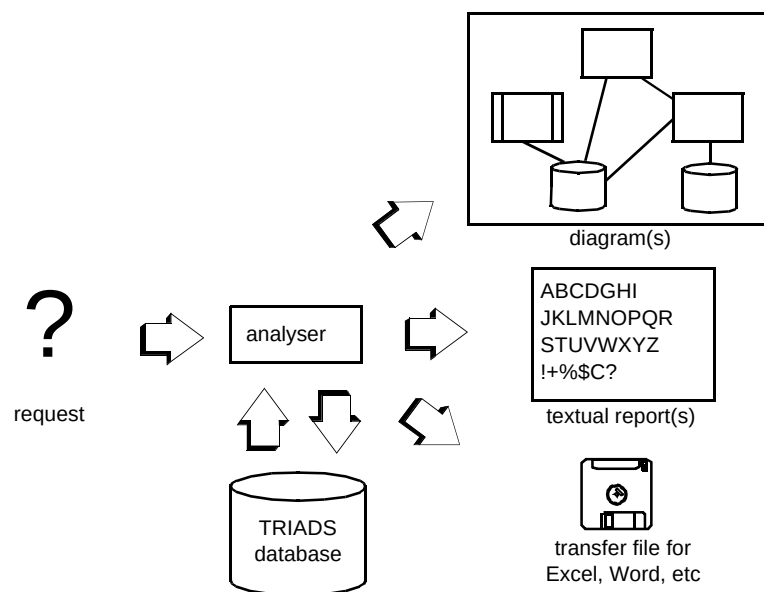


Fig 3 Schematic showing operation of the TRIADS analyser.

created diagrams, thus enabling diagram layout information and annotations to be stored.

4.3 Abstracting data

The analyser facilitates the generation of diagrams at various levels of abstraction. For example, a complex diagram showing lots of low-level detail may be abstracted to reveal its chief features and their interconnections. The analyser script would carry out the following operations:

- examine the hierarchical decomposition relationships for each instance represented by a node in the diagram (see section 5.4) and extract high-level instances at the required level of abstraction,
- derive the interconnections between the high-level instances from the interconnections shown in the low-level diagram,
- display a diagram showing all the instances and their interconnections identified in the first and second steps.

Conversely, a search may be made down the instance hierarchies to auto-generate a diagram showing decomposition detail for a high-level object.

4.4 Logical-to-physical mappings

Architectures are often expressed in both logical terms (i.e. the business processes required to carry out a task) and physical terms (i.e. how these processes are implemented by systems, people, etc). The analyser provides facilities for converting between these two viewpoints.

The logical/physical mapping facilities exploit the 'implemented by' relationship between the processes and management systems shown in Fig 1. For example, an analyser script to convert a logical viewpoint to a physical viewpoint may carry out the following operations:

- identify the physical instances associated with each process in the logical diagram, taking due account of mappings at lower levels of abstraction and filtering the result according to business context, implementation date, etc,
- create interconnections between the physical instances corresponding to the data flows between processes in the logical diagram,
- display a diagram showing all the instances, and their interconnections, identified in the first and second steps.

This approach may be used to create a first-cut physical architecture for a new logical architecture.

5. Standard viewpoints

5.1 Introduction

In addition to the analyser, TRIADS provides a number of tools for creating and editing specific viewpoints on the underlying data model. These are described in the following sections.

5.2 Hierarchical decomposition structures

Complex architectures are often modelled using multiple levels of abstraction, with instances at various abstraction levels linked via decomposition structures. This approach is particularly common for process architectures. The TRIADS hierarchy tool enables decomposition structures to be created, edited and visualised for a wide range of different entity types. The tool automatically generates a hierarchy diagram showing parent/child relationships for a selected instance (see Fig 4). Additional parent/child relationships can be created using a simple drag and drop method.

Hierarchies play a fundamental role in TRIADS. In order to build up a complete picture of the relationships for any given instance, it is often necessary to examine not just the instance itself but also its hierarchical dependants. Many TRIADS functions provide this capability. Users are encouraged to populate relationships at the lowest possible abstraction levels and use TRIADS facilities to derive higher level views.

5.3 Connections between data instances

TRIADS diagrams typically show how systems or processes interact within a particular context, but do not give a complete picture of the interactions for any individual instance. The information flows tool, by contrast, auto-generates a diagram of all the interconnections for a selected instance (see Fig 5). An information flows diagram is 'cartwheel-like' in appearance, with the instance under investigation appearing as the hub, the information flows appearing as spokes, and the connected instances occurring on the rim.

The information flows tool enables TRIADS users to rapidly visualise BT's architectures from the perspective of any individual instance. The user can readily identify how many connections the instance has, to which other systems, processes, etc, it is connected, and the nature of the data flowing across the connections. Furthermore, by use of

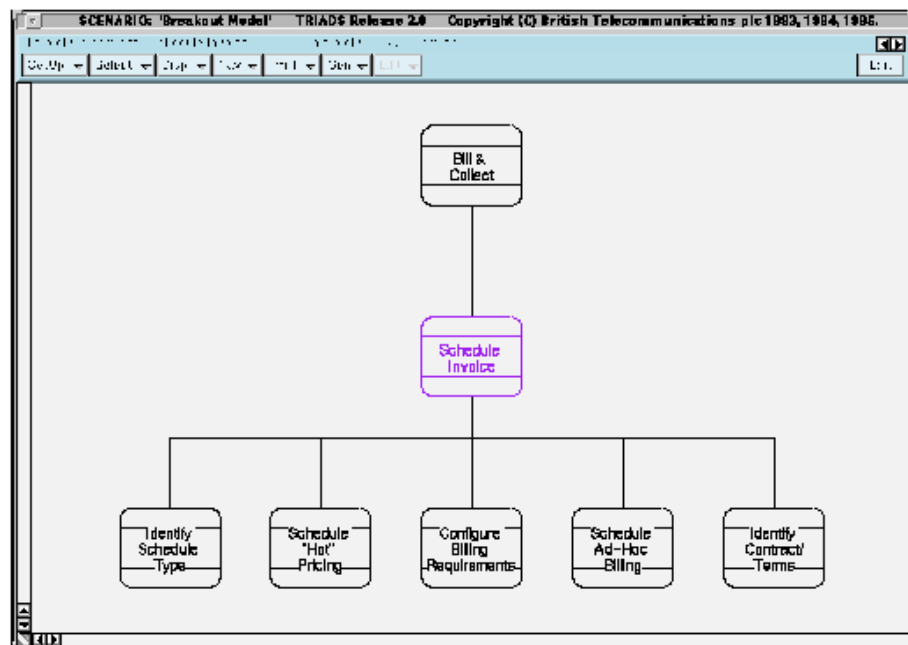


Fig 4 Example of a hierarchy diagram.

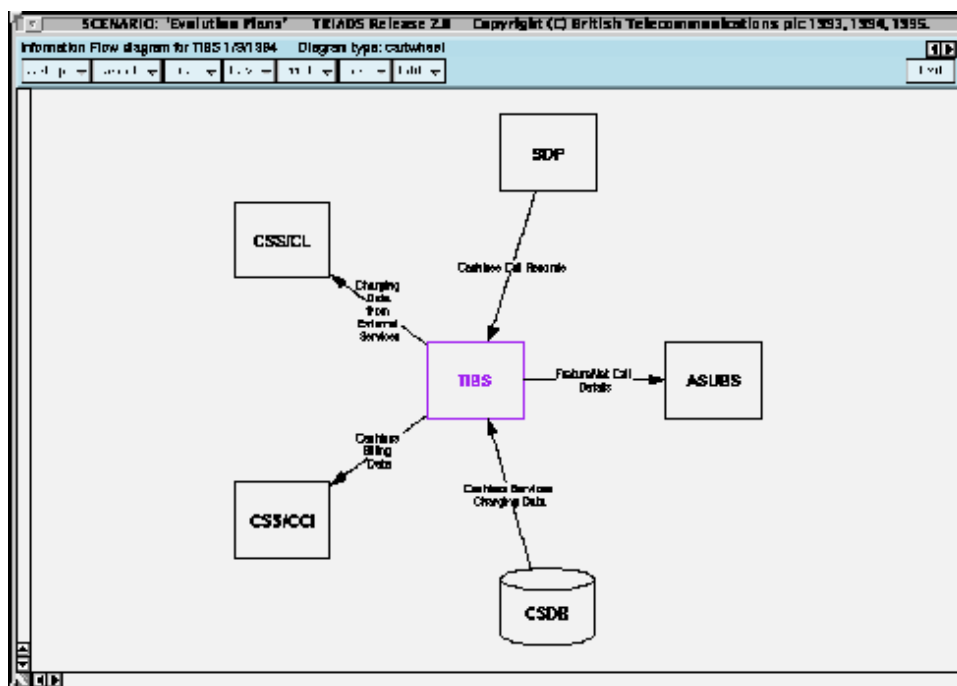


Fig 5 Example of an information flows diagram.

appropriate options, the user can examine the planned evolution of the viewpoint with time. Information flows can be created, modified or deleted to create a 'true' picture for the instance at any date.

5.4 Message sequences

Standard TRIADS diagrams tend to show the nature of the interactions between instances, but do not show the chronological sequence of messages associated with

particular operations. The message sequence tool enables this sequencing information to be acquired and displayed. The sequence is acquired by 'clicking' on a succession of links from one or more existing diagrams. The sequence information is temporarily logged in a panel, enabling it to be edited, saved to the database and, ultimately, converted into a message sequence diagram. The message sequence diagram shows the chronological sequence of the information flows for a particular operation, with time increasing down the diagram (see Fig 6). For instances which exist as children within a hierarchy (for example,

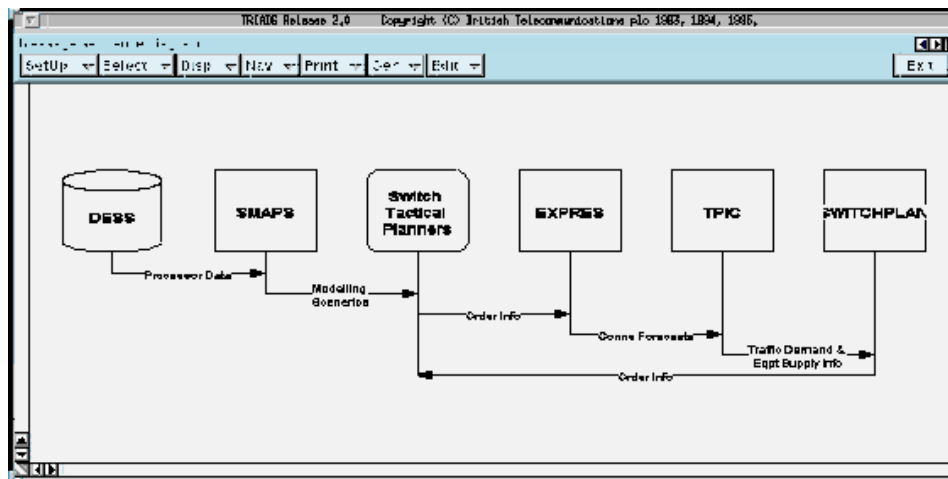


Fig 6 Example of a message sequence diagram.

processes), facilities exist to abstract complicated sequences to higher levels.

6. Migration planning

6.1 Modelling the time dimension

The primary role of architectural modelling is to provide a framework for change. Consequently, all major components of TRIADS are modelled with a time dependency and extensive facilities have been developed to support the migration planning process.

Most instances have start and end dates, indicating the periods for which they are intended to be operational within the business. In addition, each diagram is dated in such a way that it represents a snapshot of the business at a single point in time. Checks are applied during diagram creation and retrieval to ensure that the date range associated with each content instance is consistent with the diagram date. Inconsistencies are indicated by 'greying out' of nodes and links.

A migration sequence can be produced for any business viewpoint by simply creating a series of diagrams with different dates for a common set of defining instances (see section 6.4). Facilities are provided for navigating through the diagram sequence.

6.2 Timelining

The timeline tool enables evolution route maps to be created, edited and visualised for individual systems. Timeline diagrams are similar in appearance to Gantt charts, with instances plotted as horizontal bars against time (see Fig 7). The bar for each instance spans the time period between its start and end dates. Links are displayed between the bars to indicate the replacement of one instance by

another. Timeline diagrams are automatically generated from the 'replaced-by' relationships stored in the database for each instance.

Each timeline displays a critical path of predecessors and successors for a single, highlighted instance. 'Replaced-by' links can be created or deleted for the highlighted instance using simple drag-and-drop methods. The timeline tool also provides facilities for modifying the date range of a diagram, graphically updating the start and end dates for an individual instance, and displaying diagrams in a variety of different formats.

Timeline 'replaced-by' relationships provide the fundamental sequencing information required for planning the migration of any particular business viewpoint. These relationships are exploited by the interpolation tool (see section 6.4).

6.3 Comparing diagrams

The ability to view and navigate through a migration sequence is rarely sufficient. In general, migration planners are much more interested in comparing diagrams to identify similarities and differences. For a set of complex diagrams this can be an extremely difficult task. The diagram comparison tool has greatly enhanced the value of TRIADS migration planning data by enabling similarities and differences to be displayed graphically or summarised in an auto-generated report.

The diagram comparison tool can be used for any pair of TRIADS diagrams, regardless of whether they are manually or automatically generated. The tool analyses each node and link in the two diagrams and categorises it as either 'identical', 'modified' or 'different'. 'Identical' items are those which have a counterpart in the other diagram which represents the same database instance. 'Modified' items are

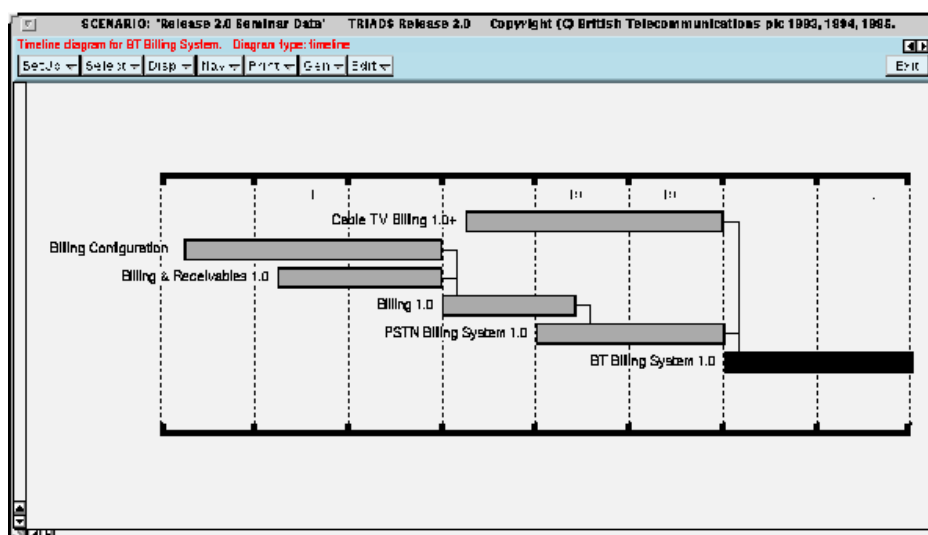


Fig 7 Example of a timeline.

those which have a counterpart in the other diagram which relates to a similar real-world object, but which is modelled by a different database instance (for example, a different version of the same OSS, or a different link between the same two processes). 'Different' items appear in only one of the two diagrams and may therefore relate to real-world objects which have been removed or introduced during the period between two migration plans. The comparison data is readily visualised by highlighting nodes and links for the various categories in different colours. Details of the comparison can be obtained by generating comparison reports for either whole diagrams or individual instances.

6.4 Creating a migration sequence

A migration planner will often have knowledge of a current situation and a distant vision and be given the task of creating a series of migration steps between the two. The interpolation tool has been provided to assist in this process.

Two existing diagrams from the same migration sequence are chosen to act as the initial and final states for the interpolation. A new intermediate diagram is then partially auto-generated by comparing the two end diagrams and automatically copying populated nodes and links to the intermediate diagram. The interpolation criteria are based on diagram comparison and timeline data. For example, if an instance exists in both end diagrams, it is assumed that it should also appear in the intermediate diagram. If an instance exists in modified form in the two diagrams, timelines are examined to determine which version of the object should be added to the intermediate diagram.

Having auto-generated part of the intermediate diagram, the interpolation tool displays the initial and final diagrams with colour coding to indicate which items have been automatically copied to the intermediate diagram. The

remainder of the intermediate diagram can then be created by drag and drop of populated symbols between diagrams or by use of the standard diagram editing facilities (see section 4.3). Colour coding of the initial and final diagrams is dynamically maintained during drag and drop to assist with housekeeping.

The interpolation tool can be used repeatedly to build up multiple steps in a migration sequence.

Use of the interpolation tool enables most routine aspects of the migration planning process to be automated. This reduces the risk of inconsistencies and enables the migration planner to concentrate on the higher value aspects of the task.

7. Data management

7.1 Scenarios

The management of architectural data requires a flexible system which enables architects and planners to dynamically create and modify data, while providing adequate access control and configuration management of the data. This balance has been achieved using a novel scheme in which the TRIADS database is partitioned into units called 'scenarios'.

Each scenario represents a unique, multi-dimensional perspective of the business. Consequently, scenarios can be used for the development of new architectural models (for example, process and systems architectures for a new service) or for carrying out speculative 'what if...?' studies. Alternative architectural models may be developed in different scenarios for comparison or impact assessment. The scenario provides a suitable partition for analysis and consistency checking of a new model, prior to submission for approval by appropriate design authorities. Authorised

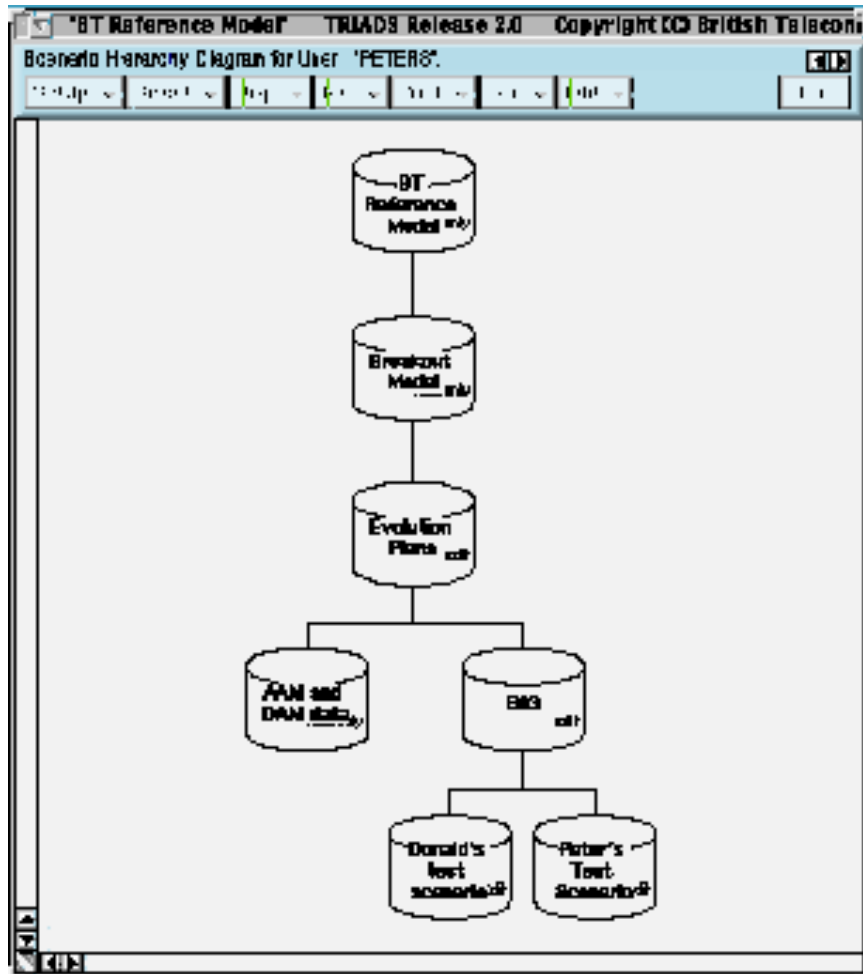


Fig 8 The TRIADS front screen, showing a typical scenario hierarchy.

scenarios may be made available to a wider community of users or incorporated into higher level models of the business.

The scenarios are arranged in a hierarchy, with the most stable, highly approved model of the business (for example, the BT Reference Model) at the top of the structure (see Fig 8). On descending the hierarchy, the stability and approval status of the scenarios tends to decrease, but the data contained in the scenarios will be more representative of latest ideas. The lowest level scenarios typically contain working models or data from speculative studies.

Each instance of data in the TRIADS database is owned by a single scenario. Visibility of instances is inherited down the hierarchy, so parent-owned instances can be viewed and reused by a child scenario, but not vice versa. Hence, each scenario is essentially a perturbation of its parent, maximising the reuse of data via inheritance. This greatly simplifies data management by avoiding the need to create multiple copies of data instances which could subsequently get out of step. The methodology ensures that all models which refer to a particular real-world entity in a

higher scenario (for example, a particular version of an OSS) refer to the same instance in the database. It is thereby guaranteed that, if any attribute of the instance is modified (for example, the date at which the new OSS version is to be deployed), the impact will immediately be knocked on to all scenarios which make use of the instance.

Any number of users may have read-only or edit access to a scenario. Consequently, scenarios can function as either private or group work areas. Typically, the high level, approved scenarios will have a large number of read-only users, but very few users with edit permissions. Low level, working scenarios will tend to be private to a single user or a small group of users, but the proportion of users with edit access is likely to be higher. Each scenario has an owner who is responsible for managing the data instances owned by the scenario. The scenario owner may assign read-only or edit access to other users.

On log-in to the TRIADS database, a user is presented with a personalised front screen showing the hierarchy of scenarios accessible from their account and the permission level (read-only, edit or owner) for each scenario. The user

selects a 'current scenario' to work in by clicking on one of the data-can symbols on the front screen (Fig 8). The current scenario then scopes the data available for viewing and editing. Data from different scenarios may be compared using multiple windows working in different current scenarios.

7.2 Management of data instances

Data instances may only be created or edited by users with edit access to the current scenario. In addition, instances may only be modified or deleted while working in the owning scenario for the instance. This ensures that the integrity of the data is maintained by preventing data instances owned by high-level scenarios (e.g. the BT Reference Model) being modified or deleted by users working in lower level scenarios.

Deletion of TRIADS instances is usually 'soft', i.e. the instances are marked as deleted, rather than being physically removed from the database. Instances which are reused in lower level scenarios are therefore flagged to the users as deleted, rather than simply disappearing from the model. A 'hard-deletion' option is also available for permanently removing unwanted data from the database.

It may sometimes be desirable for a user working in a low-level scenario to create a modified version of an instance owned higher up in the scenario hierarchy. This is particularly likely when carrying out speculative 'what if...?' studies and assessing the impact of different options. In these cases, the TRIADS application provides a 'copy' facility to enable duplicates of the instance and any of its relationships to be created in the current scenario. These duplicates are owned by the current scenario and can therefore be modified as required. TRIADS users are encouraged to exercise restraint in use of the copy facility to avoid creating a surfeit of duplicate instances and associated problems for data management.

7.3 Promotion of data instances

The status of entire architectural models or selected instances may be enhanced by promoting data up the scenario hierarchy. The promotion process transfers ownership of each data instance from a child scenario to its parent. As a result, the promoted data instances become visible to users working in the parent scenario. The child scenario retains visibility of the instances via inheritance, but editing of the instances can now only be carried out in the parent scenario. Visibility of the promoted instances is also acquired by any other scenarios which are dependants of the parent scenario. Hence, the promotion process increases the availability of the data for viewing and reuse.

Before promoting data, architectural approval is generally required from an appropriate design authority. It is up to the parent scenario owner to specify the criteria which data must satisfy before promotion can take place.

158 For low-level scenarios the approval procedures are likely

to be fairly informal. Promotion into high-level scenarios (for example, the BT Reference Model) is likely to require adherence to a strict set of rules. In particular, users will have to ensure that all specified attributes are populated and that the data set is internally consistent.

8. Implementation

The TRIADS application was developed using a suite of tools from ILOG based on the LeLisp™ variant of the Lisp programming language. The principal tools employed were Aida™, a library of graphical components, and Masai™, a graphical user interface builder. The ILOG toolset was chosen to support the evolutionary prototyping methodology used for TRIADS development.

The TRIADS application runs on a Unix™ server and is typically accessed via an X-windows emulator on client PCs. The TRIADS database is implemented on a Unix platform using an Oracle relational database management system.

9. Current status and future prospects

TRIADS is currently being used by more than 100 architects and planners within BT for high-level architectural design and migration planning. Databases are currently deployed at BT Laboratories and three sites in London. Plans are in hand to create a single logical database spanning all user sites.

TRIADS has already contributed to a large number of major programmes within BT, including Breakout OSS design, CSS migration strategy, Billing 90s, Featurenet, SMDS and Cashless. TRIADS is now the master repository for the network control layer picture-book and will shortly become the master repository for the BT systems encyclopaedia and the data architecture manual.

Until now, TRIADS has been used almost exclusively to support planning activities at the front-end of the development life cycle. However, work is currently in hand to develop a version of TRIADS suitable for more widespread use within the business. In future, TRIADS is likely to play a broader role in configuration management of system and service delivery programmes throughout the development life cycle. The architectural information will then become truly dynamic, reflecting changes in specifications, delivery dates, etc, as they occur during system design and implementation. The architectural models may also provide a basis for validation, verification and testing at the systems and networks integration level.

10. Conclusions

The TRIADS system is adding significant benefit to BT in a number of ways:

- the establishment of a common repository for architectural information with multi-user access is providing users with dynamically maintained architectural data and facilitating interworking between disparate groups across the business,
- the provision of advanced data visualisation and navigation facilities is adding value to the architectural information by enabling it to be viewed from multiple perspectives and at various levels of abstraction,
- the provision of advanced information processing and impact analysis capabilities is enabling users to fully exploit the wealth of architectural information which TRIADS has made available,
- the development of electronic tool support for migration planning is transforming the planning process.

As a result, the business is now much better equipped to manage change and respond rapidly to new threats and opportunities.

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Ian Videlo graduated from the University of Southampton with a BSc in physics. He joined BT Laboratories in 1981 where he spent several years working on the reliability of electronic and opto-electronic components.

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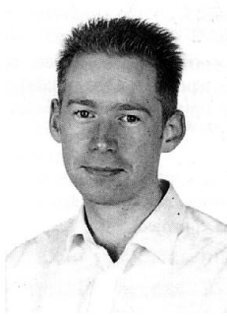
planning and management of change in complex enterprises. He is a Member of the Institute of Physics and a Chartered Physicist.



Donald Sloan graduated from the University of Strathclyde in 1984 with an honours degree in mining and petroleum engineering before going on to postgraduate study at the University of Edinburgh where he gained an MSc degree in the design and manufacture of microelectronic systems in 1985.

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